SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	**Duration / Marks:**	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
Student Name:	A Likithasri	**Reg. No.:**	192511003
Section / Batch:	Btech - CSE	**Date:**	5/10/2026

Code of Conduct

I A.Likithasri(192511003) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Likitha.A

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
- Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	**Q3**
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	**Q1, Q2**
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	**Q1, Q4**

Question 1: Two-Room Vacuum World

Topic: Agents and environments, concept of rationality | 15 minutes

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

Performance measure over a lifetime of T = 6 time steps: +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take p = 1 unless told otherwise).

- **Agent X (simple reflex):** if the current square is Dirty, Suck; otherwise move to the other square.
- **Agent Y (reflex with internal state):** if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

(a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. **[4 Marks]**

(b) Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for p = 1. **[5 Marks]**

(c) Trace Agent Y in the same way and find its total score for p = 1. **[4 Marks]**

(d) Derive the total score of X and of Y as functions of p. For which values of p is Y strictly better than X? **[3 Marks]**

(e) Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. **[4 Marks]**

Question 2: Task Environment Analysis and Agent Selection

Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes

Three AI agents are to be deployed in the following task environments:

- **E1:** An autonomous taxi driving through city traffic.
- **E2:** A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- **E3:** A chess-playing agent in a tournament game played with a clock.

Difficulty score convention: for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

Property	Easier value (0)	Harder value (1)
Observability	Fully observable	Partially observable
Number of agents	Single agent	Multi-agent
Determinism	Deterministic	Stochastic
Episodic / Sequential	Episodic	Sequential
Static / Dynamic	Static	Dynamic (semi-dynamic = 0.5)
Discrete / Continuous	Discrete	Continuous

(a) Classify each environment on all six properties with a one-phrase justification for each. **[9 Marks]**

(b) Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. **[4 Marks]**

(c) For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. **[4 Marks]**

(d) Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. **[3 Marks]**

Question 3: Turing Test Data Across the History of AI

Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

System	Era / technique	Interrogations	Correctly identified
ELIZA	1960s, keyword pattern matching	50	47
Chatbot B	2010s, scripted rules plus statistical NLP	50	36
System C	2020s, large neural language model	50	29

Reference: Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

(a) Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. **[6 Marks]**

(b) Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. **[5 Marks]**

(c) Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. **[5 Marks]**

(d) Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument). **[4 Marks]**

Question 4: Rational Route Choice under Uncertainty

Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

$$\mathbf{U = 100 - T - 20 \cdot L}$$

where T is the travel time in minutes and L = 1 if the taxi arrives after the 55-minute deadline, otherwise L = 0.

Route	Normal time (min)	Probability of disruption	Time if disrupted (min)
R1: Highway	30	0.30 (accident)	70
R2: City road	45	0.10 (signal failure)	60
R3: Ring road	50	0 (never disrupted)	50

(a) Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. **[4 Marks]**

(b) Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose. **[5 Marks]**

(c) A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. **[4 Marks]**

(d) Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. **[4 Marks]**

(e) Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. **[3 Marks]**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Problem Understanding	**20%**	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	**20%**	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	**25%**	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	**20%**	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	**15%**	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria	Problem Understanding (4)	Method Selection (4)	Solution Process (5)	Accuracy of Calculation (4)	Interpretation of Results (3)	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Question 1: Two-Room Vacuum World

A vacuum-cleaner agent operates in a world containing two squares, **A and B**. Initially, both squares are dirty, and the agent starts in square **A**. The agent can perform four actions: **Suck, Left, Right, and NoOp**.

Once a square is cleaned, it remains clean.

The performance measure is calculated over **6 time steps**:

- **+1** for every clean square at the end of each time step.
- **−p** for every Left or Right movement.
- For the tracing questions, **p = 1**.

Agent X is a **simple reflex agent**, while Agent Y is a **reflex agent with internal memory**.

(a) PEAS Description

PEAS stands for **Performance Measure, Environment, Actuators, and Sensors**.

Performance Measure

The agent receives:

- **+1** for each clean square at the end of every time step.
- A penalty of **−p** for every Left or Right movement.

The objective is to **maximize the total performance score**.

Environment

The environment consists of:

- Two squares: **A and B**
- Both squares are initially dirty.
- A cleaned square remains clean permanently.
- The agent starts in square **A**.

Actuators

The agent can perform the following actions:

- **Suck** – cleans the current square.
- **Right** – moves from A to B.
- **Left** – moves from B to A.
- **NoOp** – performs no action.

Sensors

The sensors provide information about:

- The agent's current location.
- Whether the current square is **Dirty or Clean**.

(b) Trace of Agent X

Rule of Agent X

Agent X follows a simple reflex rule:

- If the current square is **Dirty**, perform **Suck**.
- If the current square is **Clean**, move to the other square.

Since **p = 1**, every movement produces a penalty of **−1**.

Step-by-Step Trace

Time Step	Location	Percept	Action	Clean Squares	Movement Penalty	Step Score
1	A	A is Dirty	Suck	A	0	1
2	A	A is Clean	Right	A	−1	0
3	B	B is Dirty	Suck	A, B	0	2
4	B	B is Clean	Left	A, B	−1	1
5	A	A is Clean	Right	A, B	−1	1
6	B	B is Clean	Left	A, B	−1	1

Total Score of Agent X

After time step 3, both squares are already clean. However, Agent X does not remember this and continues moving between the two squares.

Therefore,

Total Score of Agent X

= 1 + 0 + 2 + 1 + 1 + 1

= 6

(c) Trace of Agent Y

Rule of Agent Y

Agent Y uses both the current percept and its internal memory.

- If the current square is **Dirty**, perform **Suck**.
- If the current square is **Clean** and both squares have not yet been confirmed clean, move to the other square.
- If both squares are known to be clean, perform **NoOp**.

Step-by-Step Trace

Time Step	Location	Percept	Memory	Action	Clean Squares	Movement Penalty	Step Score
1	A	A is Dirty	A not clean	Suck	A	0	1
2	A	A is Clean	B not checked	Right	A	−1	0
3	B	B is Dirty	A is clean	Suck	A, B	0	2
4	B	B is Clean	Both clean	NoOp	A, B	0	2
5	B	B is Clean	Both clean	NoOp	A, B	0	2
6	B	B is Clean	Both clean	NoOp	A, B	0	2

Total Score of Agent Y

Agent Y stops moving after both squares have been cleaned, so it avoids unnecessary movement penalties.

Therefore,

Total Score of Agent Y

= 1 + 0 + 2 + 2 + 2 + 2

= 9

(d) Total Score as a Function of p

Score of Agent X

Agent X makes four movements:

- Step 2 → Right
- Step 4 → Left
- Step 5 → Right
- Step 6 → Left

Therefore, the total movement penalty is:

−4p

The clean-square rewards are:

1 + 1 + 2 + 2 + 2 + 2 = 10

Hence,

Score(X) = 10 − 4p

Score of Agent Y

Agent Y makes only one movement:

- Step 2 → Right

Therefore, the total movement penalty is:

−p

The clean-square rewards are:

1 + 1 + 2 + 2 + 2 + 2 = 10

Hence,

Score(Y) = 10 − p

Comparison of the Two Agents

For Agent Y to have a strictly higher score:

Score(Y) > Score(X)

Substituting the equations:

10 − p > 10 − 4p

−p > −4p

3p > 0

Therefore,

p > 0

Thus, whenever the movement penalty is positive, Agent Y obtains a higher score.

If:

p = 0

both agents obtain the same score.

For p = 1

Score(X) = 10 − 4(1) = 6

Score(Y) = 10 − 1 = 9

Therefore, Agent Y obtains a higher score.

(e) Rationality and Interpretation

Rational Agent

Agent Y is the **rational agent** under the given performance measure.

A rational agent chooses actions that maximize its expected performance based on the available

information.
Agent Y uses:

- The current percept.
- Previous percept history.
- Internal memory.
- The performance measure.

Why Agent X Cannot Match Agent Y

Agent X is a simple reflex agent. It considers only the current condition of the square.
After both squares have been cleaned, Agent X sees:
Current square = Clean
Its rule tells it to move to the other square.
Therefore, it continues moving between the two squares.
These movements do not clean any additional square and only produce penalties.
Agent Y, however, remembers that both **A and B** have already been cleaned. Therefore, it selects:
NoOp
and avoids unnecessary movement.

Comparison

Agent X	**Agent Y**
Uses only the current percept	Uses current percept and memory
Does not remember previous cleaning	Remembers that both squares are clean
Continues moving after cleaning both squares	Performs NoOp after both squares are clean
Makes 4 movements	Makes 1 movement
For p = 1, score = 6	For p = 1, score = 9

Conclusion

The internal memory of Agent Y allows it to make better decisions using the history of percepts. Once both squares are clean, it recognizes that further movement is unnecessary and chooses **NoOp**.
This reduces the movement penalty and gives a higher overall performance.
Therefore, for any **positive movement penalty (p > 0)**, Agent Y achieves the higher performance because it uses percept history and internal state to select its actions.

Question 2: Task Environment Analysis and Agent Selection

Given Environments

- **E1:** Autonomous taxi driving through city traffic
- **E2:** Part-picking robot using a camera to identify defective parts on a moving conveyor belt
- **E3:** Chess-playing agent competing in a timed tournament

For each environment, the six properties are classified using the values given in the question.

(a) Classification of the Environments

E1 – Autonomous Taxi

Property	Classification	Reason
Observability	Partially observable	The taxi cannot obtain complete information about all traffic conditions.
Number of agents	Multi-agent	Other vehicles, pedestrians, and road users also influence the situation.
Determinism	Stochastic	The behaviour of traffic participants cannot be predicted exactly.
Episodic / Sequential	Sequential	A driving decision can affect the situations encountered later.
Static / Dynamic	Dynamic	Road and traffic conditions can change while the taxi is operating.
Discrete / Continuous	Continuous	Variables such as speed and position change continuously.

E2 – Part-Picking Robot

Property	Classification	Reason
Observability	Partially observable	The camera may not identify every defect or condition perfectly.
Number of agents	Single-agent	The robot carries out the picking task independently.
Determinism	Stochastic	Detection and movement of parts can involve uncertainty.
Episodic / Sequential	Sequential	Decisions made during picking affect subsequent actions.
Static / Dynamic	Dynamic	The parts continue moving along the conveyor.
Discrete / Continuous	Continuous	The robot and conveyor involve continuous movement.

E3 – Chess-Playing Agent

Property	Classification	Reason
Observability	Fully observable	The positions of the pieces on the chessboard are visible.
Number of agents	Multi-agent	Two players are competing against each other.

Property	Classification	Reason
Determinism	Stochastic	The opponent's next move cannot be known in advance.
Episodic / Sequential	Sequential	Each move influences the possible moves that follow.
Static / Dynamic	Semi-dynamic	The board does not change by itself, but the game clock continues running.
Discrete / Continuous	Discrete	Chess moves and board states are represented as distinct states.

(b) Difficulty Score

According to the given scoring rule:

- Harder value = **1**
- Easier value = **0**
- Semi-dynamic = **0.5**

E1 – Autonomous Taxi

- Partially observable = 1
- Multi-agent = 1
- Stochastic = 1
- Sequential = 1
- Dynamic = 1
- Continuous = 1

Difficulty score = 1 + 1 + 1 + 1 + 1 + 1 = 6

E2 – Part-Picking Robot

- Partially observable = 1
- Single-agent = 0
- Stochastic = 1
- Sequential = 1
- Dynamic = 1
- Continuous = 1

Difficulty score = 1 + 0 + 1 + 1 + 1 + 1 = 5

E3 – Chess Agent

- Fully observable = 0
- Multi-agent = 1
- Stochastic = 1
- Sequential = 1
- Semi-dynamic = 0.5
- Discrete = 0

Difficulty score = 0 + 1 + 1 + 1 + 0.5 + 0 = 3.5

Ranking from Hardest to Easiest

Rank	Environment	Difficulty Score
1	E1 – Autonomous Taxi	**6**
2	E2 – Part-Picking Robot	**5**
3	E3 – Chess Agent	**3.5**

Therefore, **E1 is the hardest environment**, while **E3 is the easiest according to the given scoring sys.**

(c) Suitable Agent Architecture

Hardest Environment – E1: Autonomous Taxi

A **utility-based agent** is suitable for the autonomous taxi.

The taxi has to make decisions by considering several factors at the same time, such as:

- Safety
- Travel time
- Traffic conditions
- Route selection

A utility-based architecture can compare different possible actions and select the one that gives the best overall outcome.

Easiest Environment – E3: Chess Agent

A **utility-based agent** can also be used for the chess environment.

The agent can evaluate possible chess moves and select the move that provides the greatest expected advantage. This is appropriate because chess is multi-agent and sequential, so the agent must consider how its current move affects future positions.

(d) Interpretation

The complexity of an agent is mainly influenced by the properties of its environment rather than simply by the name of the application.

- **Partial observability** means the agent must work with incomplete information.
- **Multi-agent environments** require consideration of the actions of other agents.
- **Stochastic environments** require the agent to handle uncertain outcomes.
- **Sequential environments** require decisions to account for future situations.
- **Dynamic environments** require the agent to respond to changes while operating.
- **Continuous environments** require handling continuously changing values such as position and speed.

Thus, the environment classification helps determine how much information, reasoning, and decision-making capability the agent program needs.

Conclusion

Among the three environments, **E1 – Autonomous Taxi** has the highest difficulty score of **6**, followed by **E2 – Part-Picking Robot** with **5**, and **E3 – Chess Agent** with **3.5**. The taxi environment is the most demanding because it combines partial observability, multiple agents, uncertainty, sequential decisions, dynamic changes, and continuous variables. Therefore, the properties of the environment should guide the selection and complexity of the AI agent architecture.

Question 3: Turing Test Data Across the History of AI

Given Data

Each AI system was evaluated in **50 separate interrogations**. The number of times the judges correctly identified the system as a machine is:

System	Correctly Identified as Machine
ELIZA	47
Chatbot B	36
System C	29

The cases in which the judge did not identify the machine correctly are considered **fooling cases**.

(a) Correct-Identification Rate and Fooling Rate

ELIZA

ELIZA was recognized as a machine in 47 out of 50 cases.

Correct-identification rate:

$47/50 \times 100 = \mathbf{94\%}$

Number of times it fooled the judges:

$50 - 47 = \mathbf{3}$

Fooling rate:

$3/50 \times 100 = \mathbf{6\%}$

Therefore, **ELIZA was correctly identified 94% of the time and fooled the judges 6% of the time.**

Chatbot B

Chatbot B was correctly recognized 36 times.

Correct-identification rate:

$36/50 \times 100 = \mathbf{72\%}$

Fooling cases:

$50 - 36 = \mathbf{14}$

Fooling rate:

$14/50 \times 100 = \mathbf{28\%}$

Therefore, **Chatbot B had a 72% correct-identification rate and a 28% fooling rate.**

System C

System C was correctly recognized as a machine 29 times.

Correct-identification rate:

$29/50 \times 100 = \mathbf{58\%}$

Fooling cases:

$50 - 29 = \mathbf{21}$

Fooling rate:

$21/50 \times 100 = \mathbf{42\%}$

Therefore, **System C had a 58% correct-identification rate and fooled the judges in 42% of the cases.**

Comparison

System	Correct-Identification Rate	Fooling Rate
ELIZA	94%	6%
Chatbot B	72%	28%
System C	58%	42%

System C has the **highest fooling rate**, which means it was the most successful at being mistaken for a human.

(b) Checking Turing's 70% Criterion

The given criterion states that the machine should be correctly identified **no more than 70% of the time**.

Therefore:

ELIZA

94% > 70%

So, **ELIZA does not satisfy the criterion.**

Its difference from the 50% guessing level is:

94% − 50% = **44 percentage points**

Chatbot B

72% > 70%

So, **Chatbot B does not satisfy the criterion.**

Its difference from the 50% level is:

72% − 50% = **22 percentage points**

System C

58% < 70%

So, **System C satisfies the criterion.**

Its difference from the 50% level is:

58% − 50% = **8 percentage points**

System	Correct-Identification Rate	70% Criterion	Difference from 50%
ELIZA	94%	Not satisfied	44 points
Chatbot B	72%	Not satisfied	22 points
System C	58%	Satisfied	8 points

Hence, **System C is closest to the 50% guessing level and is the only system that meets the given 70% condition.**

(c) Change in Fooling Rate

ELIZA to System C

Fooling rate of ELIZA = 6%

Fooling rate of System C = 42%

Increase:

42% − 6% = **36 percentage points**

To compare them as a ratio:

42/6 = **7**

Thus, the fooling rate of System C is **7 times** that of ELIZA.

Chatbot B to System C

Fooling rate of Chatbot B = 28%
Fooling rate of System C = 42%
Increase:
42% − 28% = **14 percentage points**
Ratio:
42/28 = **1.5**
Thus, System C's fooling rate is **1.5 times** the fooling rate of Chatbot B.

(d) Interpretation
The results indicate that the later system, **System C**, was more successful in producing responses that appeared human to the judges. Its fooling rate of **42%** is much higher than ELIZA's 6% and Chatbot B's 28%.
However, the Turing Test is concerned with **human-like behaviour in conversation**. A high fooling rate does not by itself prove that the AI actually understands the language, thinks, or reasons in the same way as a human.
This can be related to the **Chinese Room argument**, which explains that a system may produce appropriate responses by following rules or patterns without necessarily understanding the meaning of those responses.
Therefore, System C's result shows stronger **human-like conversational behaviour**, but it should not be taken as direct evidence of genuine understanding or human-like thinking.

Conclusion
The fooling rate improved from **6% for ELIZA**, to **28% for Chatbot B**, and finally to **42% for System C**. System C therefore performed best in the given Turing Test data and was the only system to satisfy the **70% criterion**.
Overall, the results show improvement in the ability of AI systems to appear human during conversation. However, passing or performing well in the Turing Test does not necessarily establish genuine understanding or reasoning.

Question 4: Rational Route Choice under Uncertainty

A taxi needs to reach the airport within a deadline of 55 minutes. The utility of each route is calculated using:

U = 100 − T − 20L

where:

- **T** = travel time
- **L = 1** if the taxi reaches after 55 minutes
- **L = 0** if the taxi reaches within 55 minutes

The available routes are:

Route	Normal Time	Disruption Probability	Time if Disrupted
R1 – Highway	30 min	0.30	70 min
R2 – City Road	45 min	0.10	60 min
R3 – Ring Road	50 min	0	50 min

(a) Calculate the Utility for Each Route

Route R1 – Highway

Under normal conditions, the travel time is 30 minutes, which is within the 55-minute deadline.

L = 0

U = 100 − 30 − 20(0)

U = 70

If the highway is disrupted, the travel time becomes 70 minutes. This is beyond the deadline, so L = 1.

U = 100 − 70 − 20(1)

U = 10

Therefore, R1 has a utility of **70 normally** and **10 when disrupted**.

Route R2 – City Road

Normally, the travel time is 45 minutes, which is within the deadline.

L = 0

U = 100 − 45 − 20(0)

U = 55

During disruption, the travel time becomes 60 minutes, which exceeds the deadline.

L = 1

U = 100 − 60 − 20(1)

U = 20

Therefore, R2 has a utility of **55 normally** and **20 when disrupted**.

Route R3 – Ring Road

R3 has no disruption and takes 50 minutes, which is within the deadline.

L = 0

U = 100 − 50 − 20(0)

U = 50

Utility Summary

Route	Normal Utility	Disrupted Utility
R1	70	10
R2	55	20

Route	Normal Utility	Disrupted Utility
R3	50	50

(b) Calculate the Expected Utility

The expected utility is calculated by considering the probability of each possible condition.

Route R1

Probability of normal condition = 0.70

Probability of disruption = 0.30

Expected Utility:

EU(R1) = (0.70 × 70) + (0.30 × 10)

= 49 + 3

EU(R1) = 52

Route R2

Probability of normal condition = 0.90

Probability of disruption = 0.10

Expected Utility:

EU(R2) = (0.90 × 55) + (0.10 × 20)

= 49.5 + 2

EU(R2) = 51.5

Route R3

There is no disruption, so:

EU(R3) = 1.00 × 50

EU(R3) = 50

Comparison

Route	Expected Utility
R1	**52**
R2	51.5
R3	50

Since **R1 has the highest expected utility of 52**, the rational choice is **Route R1 (Highway)**.

(c) Effect of the Radio Alert

The radio alert increases the probability of disruption on R1 from 0.30 to **0.80**.

Therefore:

Probability of normal condition = 0.20

Probability of disruption = 0.80

For R1:

EU(R1) = (0.20 × 70) + (0.80 × 10)

= 14 + 8

EU(R1) = 22

The expected utilities of the other routes remain unchanged:

- R2 = **51.5**
- R3 = **50**

Therefore:

Route	New Expected Utility
R1	22
R2	**51.5**
R3	50

After receiving the radio alert, **R2 becomes the best choice** because it now has the highest expected utility.

(d) If R1 Is Selected and a Disruption Occurs

In part (b), R1 was selected because it had the highest expected utility:

EU(R1) = 52

If a disruption actually occurs, the taxi takes 70 minutes and reaches after the deadline.

The actual utility becomes:

$U = 100 - 70 - 20(1)$

U = 10

This does not mean that choosing R1 was irrational. The decision was made using the probability of disruption available at that time. The agent could not know in advance whether the disruption would actually occur.

A rational agent chooses the action with the highest expected performance based on the information available to it.

(e) Effect of New Information on the Rational Decision

Initially, R1 had a disruption probability of only 30%, so its expected utility was the highest at **52**. Therefore, R1 was selected.

After the radio alert, the disruption probability for R1 increased to 80%. Its expected utility then decreased to **22**.

As a result, R2 became the better option with an expected utility of **51.5**.

This shows that a rational agent can change its decision when it receives new information. The agent does not simply follow one fixed route; instead, it updates its choice according to the current information and expected outcomes.

Conclusion

Before receiving the radio alert, **R1 (Highway)** is the rational choice because it has the highest expected utility of **52**. After the alert increases its disruption probability, **R2 (City Road)** becomes the preferred route with an expected utility of **51.5**.

Thus, rational decision-making under uncertainty involves considering probabilities, calculating expected utility, and changing the action when new information changes the expected outcome.